



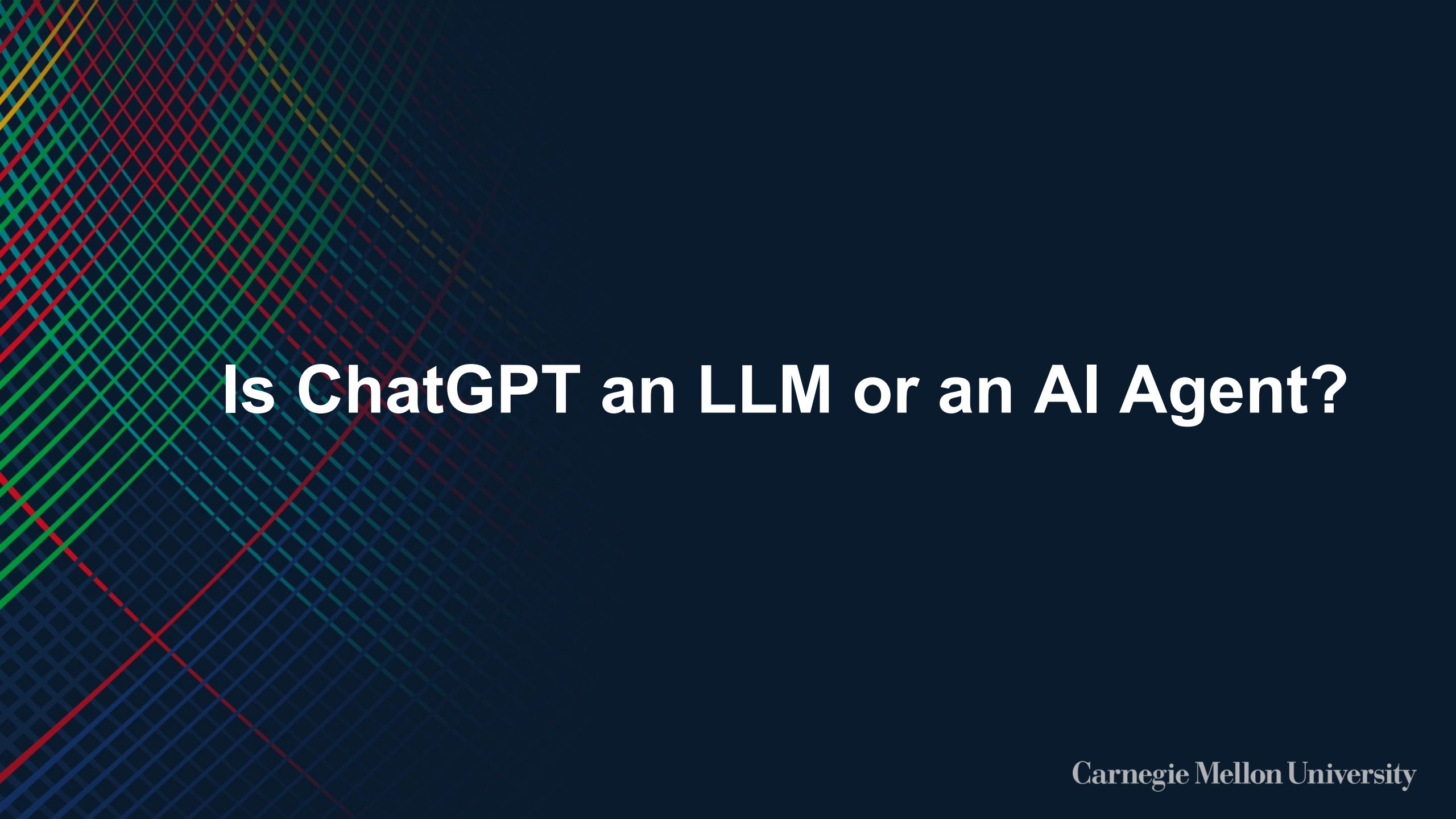
Lecture 3: Introduction to Agentic Engineering

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Agenda

- LLMs vs Agents
- LLM Limitations
- LLM Prompt Types
- Anatomy of AI Agents
- Skills vs Tools
- Additional Resources



Is ChatGPT an LLM or an AI Agent?

ChatGPT Back in the Day!

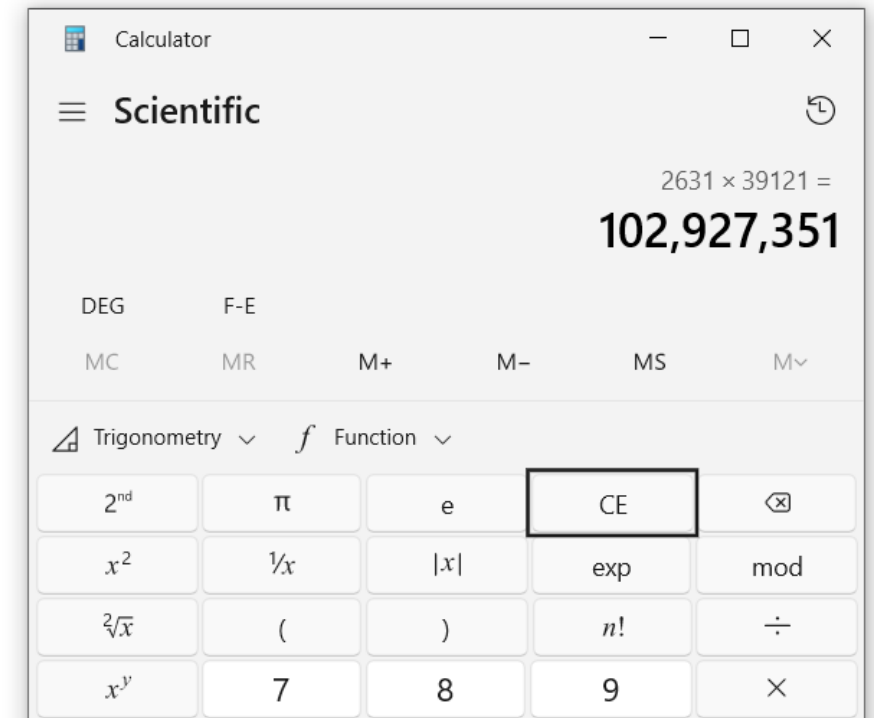
ChatGPT 3.5 ▾

You
What is the result of 26 multiplied by 25?

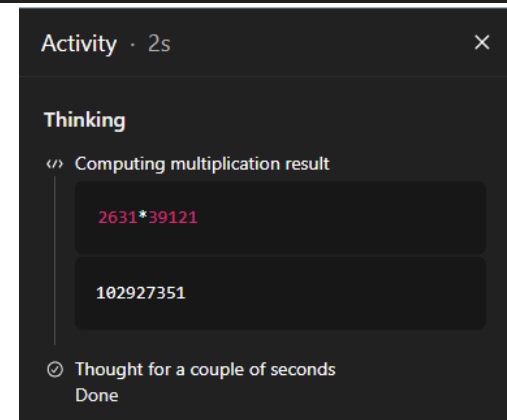
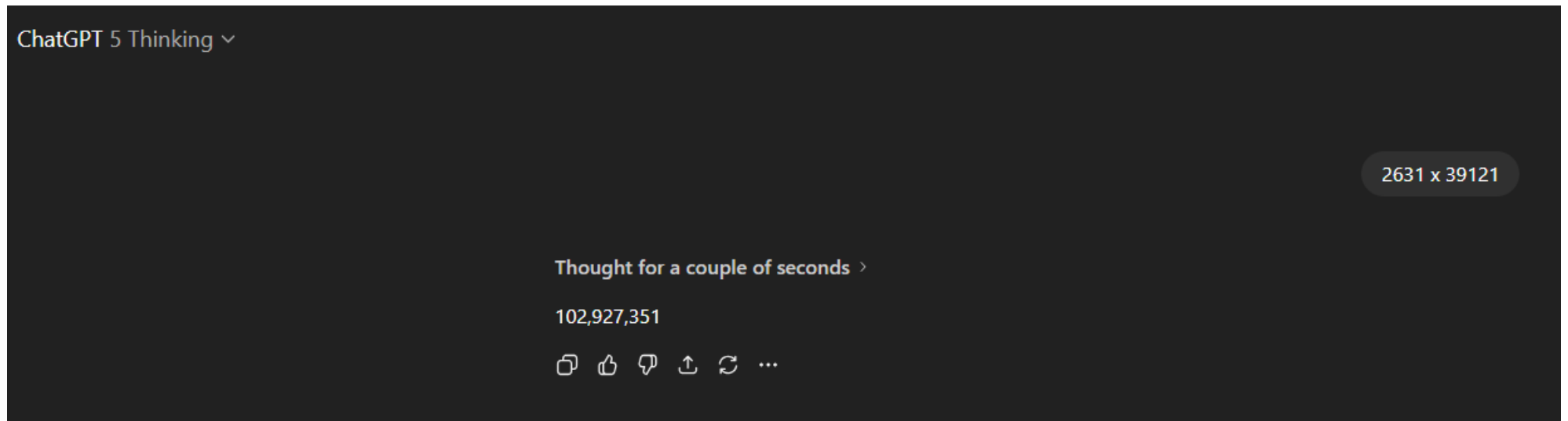
ChatGPT
26 multiplied by 25 equals 650.

You
What is the result of 2631 multiplied by 39121?

ChatGPT
The result of 2631 multiplied by 39121 is 103,034,151.
📄 👍 ↻



ChatGPT 5 – In Action

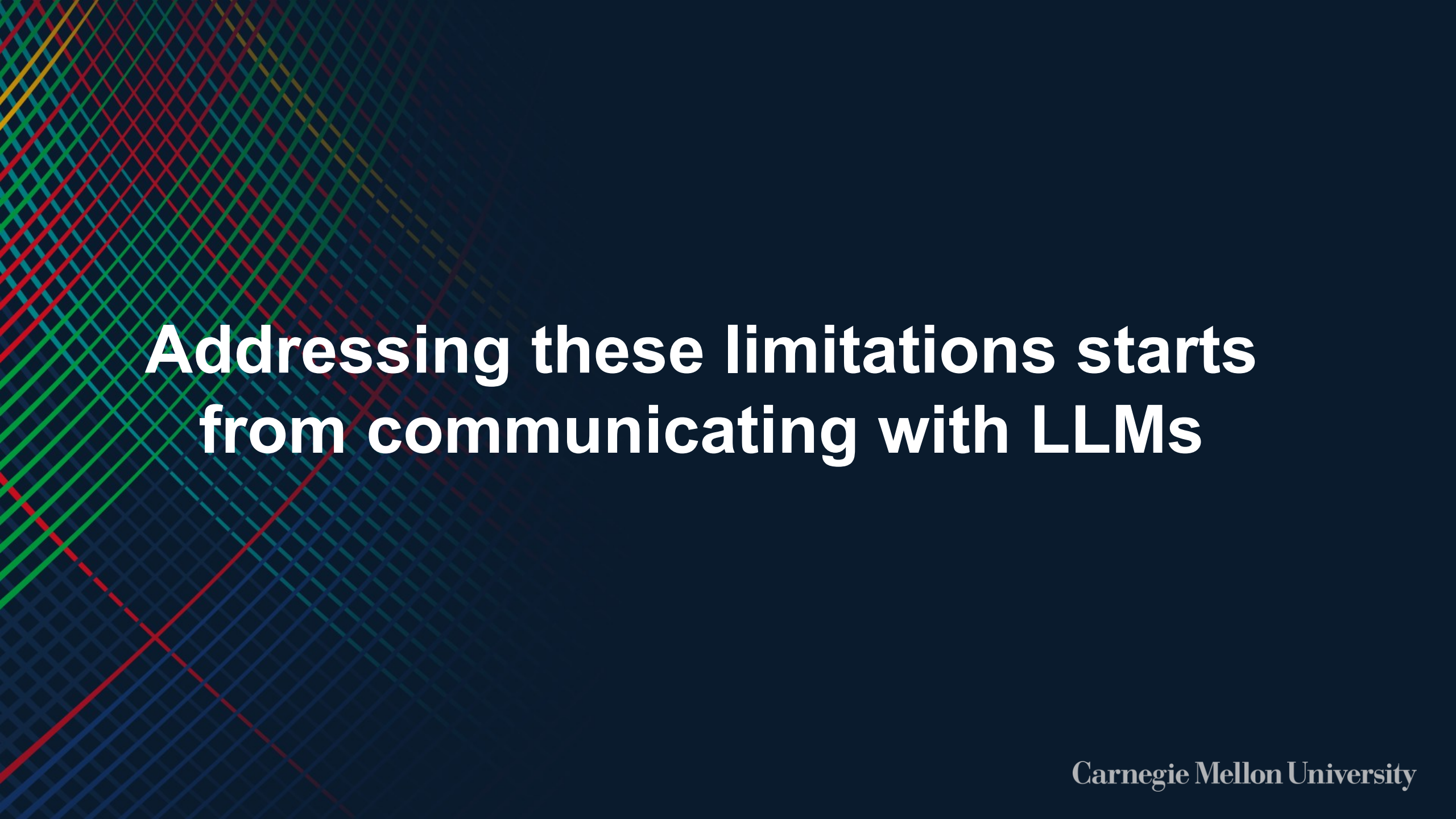




LLM Limitations

LLMs are not perfect! Keep in mind the following limitations with LLMs:

- LLMs don't work well with tabular data and mathematical calculations.
- Length of Input and Output of LLMs is limited, implying limited data to be provided and potentially limited output to be consumed.
- Knowledge Cutoff: LLMs are limited to the training date.
- Inability to take actions.
- Bias and Toxicity.
- Hallucinations!



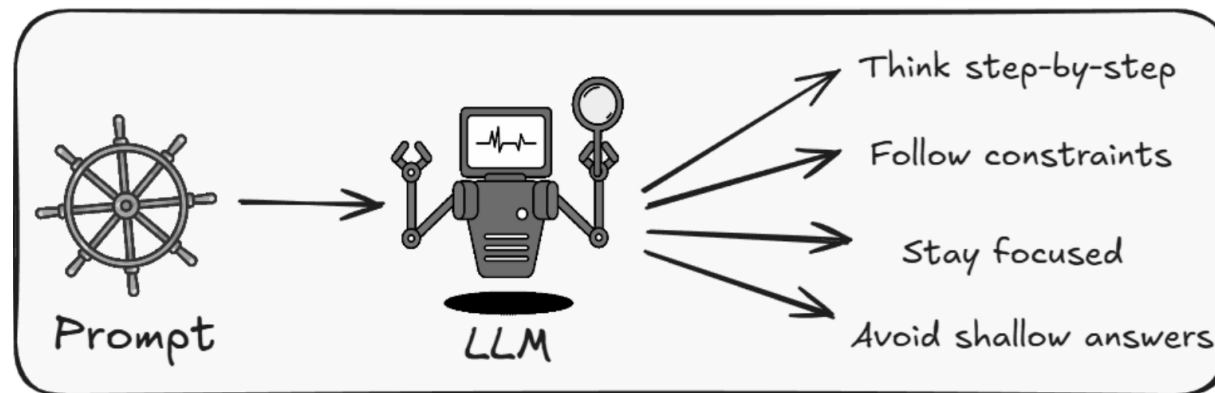
**Addressing these limitations starts
from communicating with LLMs**

How to communicate with LLMs?

LLMs don't automatically know what you want!

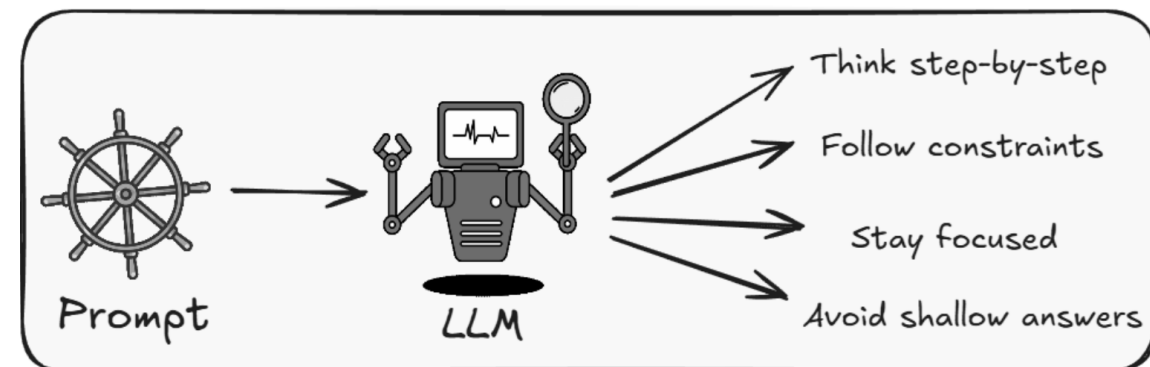
With prompts, you are not changing the model weights (the learned parameters inside the model).

Instead, you are changing the instructions, and this changes everything.



Overcoming LLM Limitations: Prompt Engineering

- Prompt engineering refers to the systematic design of effective prompts for interacting with AI models.
- Prompt Engineering requires meticulously designing prompts to include essential context, guiding the language model towards more accurate and relevant responses without changing model weights or parameters.
- Bad Prompt = Bad Output.



PROMPTING AN LLM IS LIKE TALKING TO A VERY CONFIDENT GENIE.

VAGUE PROMPT



PRECISE PROMPT





So ..

How to make LLM prompts effective?



LLM Prompt Components

1. **Task:** What you want the AI to do (for example: write, summarize, or generate something).
2. **Format:** How the answer should look (length, type like an email or blog post, file type, etc.).
3. **Topic:** What the request is about? What is the main subject?
4. **Tone or style:** How it should sound (serious, casual, professional, friendly, etc.)?
5. **Context:** Extra background or details that help the AI understand the situation or audience.
6. **Requirements or constraints:** Rules the AI must follow, including what to include or avoid.

LLM Prompt Components – Cont'd

7. **Goal:** Tell the AI what outcome you want. what you want the reader to think, feel, or do.
 - For example, convince the reader or get them to take action.
8. **Role:** Tell the AI who it should act as or write from the perspective of.
 - For example, a teacher, an expert, a student, or a specific type of audience.



LLM Prompt Example

Act as a professional career coach and write a short follow-up email about a recent job interview. The email should be professional and confident, no longer than 150 words, and written in a polite and positive tone. The context is that the interview happened two days ago, and the candidate wants to thank the interviewer and restate interest in the position. The goal is to encourage the hiring manager to move forward with the candidate. Do not mention salary or benefits, avoid sounding desperate, and end with a clear but polite call to action.



LLM Prompt Example

Act as a professional career coach and write a short follow-up email about a recent job interview. The email should be professional and confident, no longer than 150 words, and written in a polite and positive tone. The context is that the interview happened two days ago, and the candidate wants to thank the interviewer and restate interest in the position. The goal is to encourage the hiring manager to move forward with the candidate. Do not mention salary or benefits, avoid sounding desperate, and end with a clear but polite call to action.

**But .. This prompt is unstructured ...
How would we optimize it?**



LLM Prompt Example using Structured Prompting

#Role: You are a professional career coach.

#Task: Write a short email.

#Topic: Following up after a job interview.

#Format: A professional email, no more than 150 words.

#Tone / Style: Polite, confident, and professional.

#Context: The candidate interviewed two days ago and wants to express appreciation and restate interest in the role.

#Goal: Encourage the hiring manager to move forward with the candidate.

#Requirements / Constraints:

- Do not sound desperate.
- Do not mention salary or benefits.
- End with a clear but polite call to action.



Is Structured Prompting Effective?

Research shows that using structured prompts can improve the LLM output accuracy by 10-20%.

Refer to the following URLs:

- <https://arxiv.org/abs/2402.11770>
- <https://arxiv.org/abs/2305.06599>
- <https://arxiv.org/abs/2506.16123>

Is Structured Prompting Effective? – Cont'd

Researchers from University of Pennsylvania found that the structure and formatting of prompts can significantly affect LLM performance. Specifically, well-structured prompts often outperform unformatted ones.

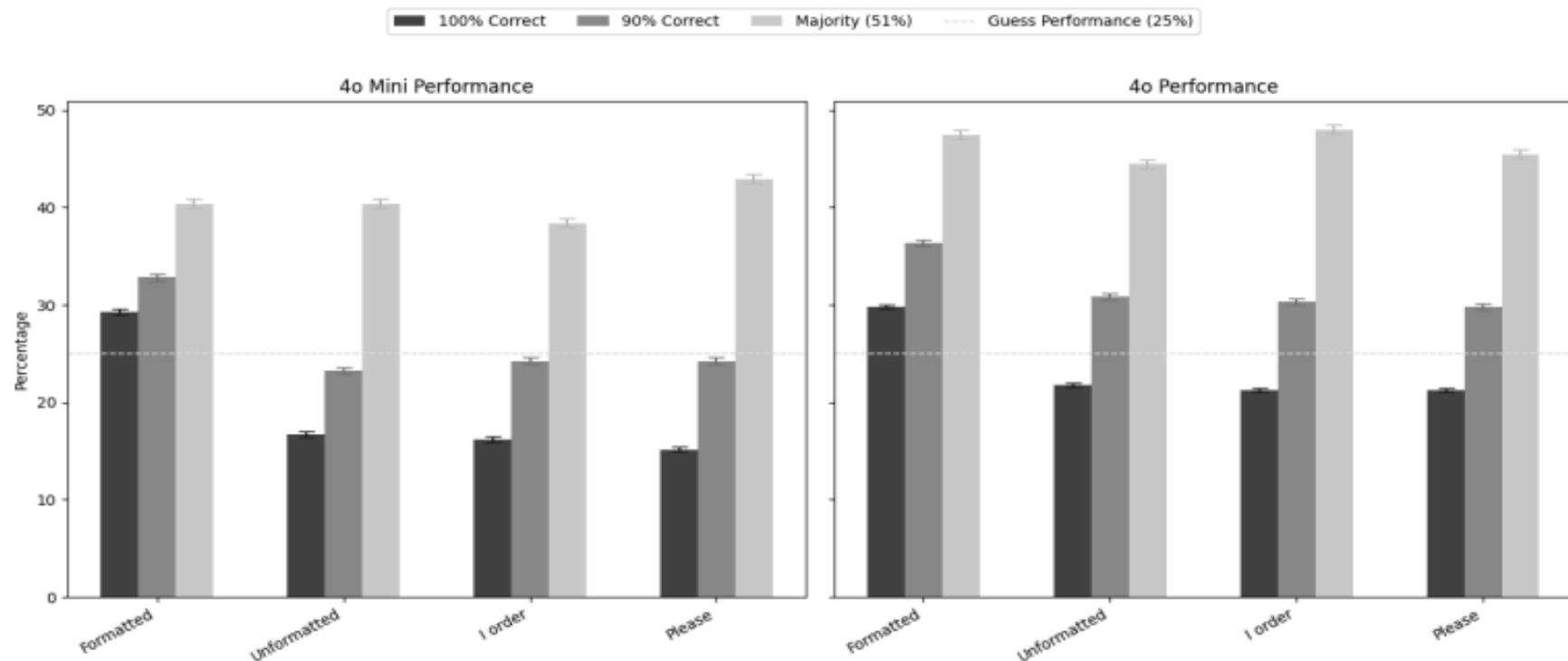


Figure 1 from Meincke, Lennart and Mollick, Ethan R. and Mollick, Lilach and Shapiro, Dan, Prompting Science Report 1: Prompt Engineering is Complicated and Contingent (March 04, 2025). Available at SSRN: <https://ssrn.com/abstract=5165270>

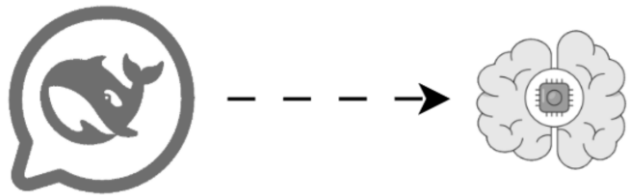


LLM Limitations - Revisit

LLMs are not perfect! Some LLM limitations include:

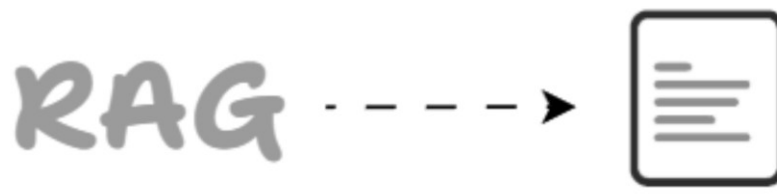
- **Knowledge Cutoff:** LLMs are bound by their training date. RAG systems are used to address this limitation.
- **Inability to take actions:** Large language models are not equipped to execute interactive tasks such as searches, calculations, or data retrievals, which considerably restricts their capabilities. **AI Agents are used to address this limitation!**

LLMs, RAGs and AI Agents!



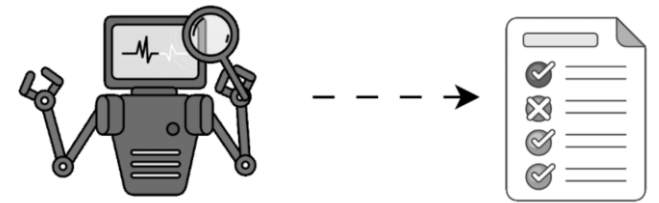
LLM is smart but static

An LLM can't call APIs, access the web, or fetch information on its own.



Retrieves fresh knowledge

RAG improves an LLM by retrieving external documents and inputting them into the LLM as context to generate a response.



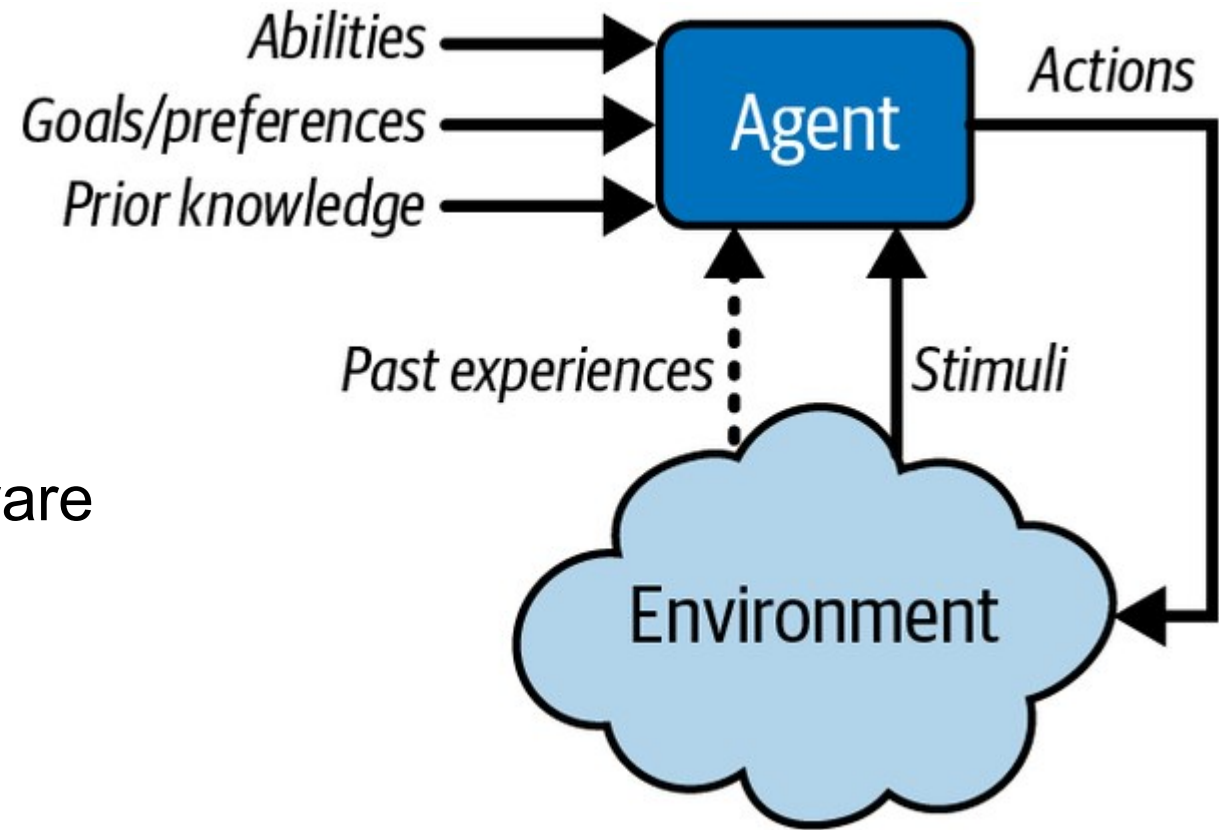
Agent thinks and act

AI Agent uses an LLM, invokes tools, orchestrates workflows, and makes decisions just like a real assistant

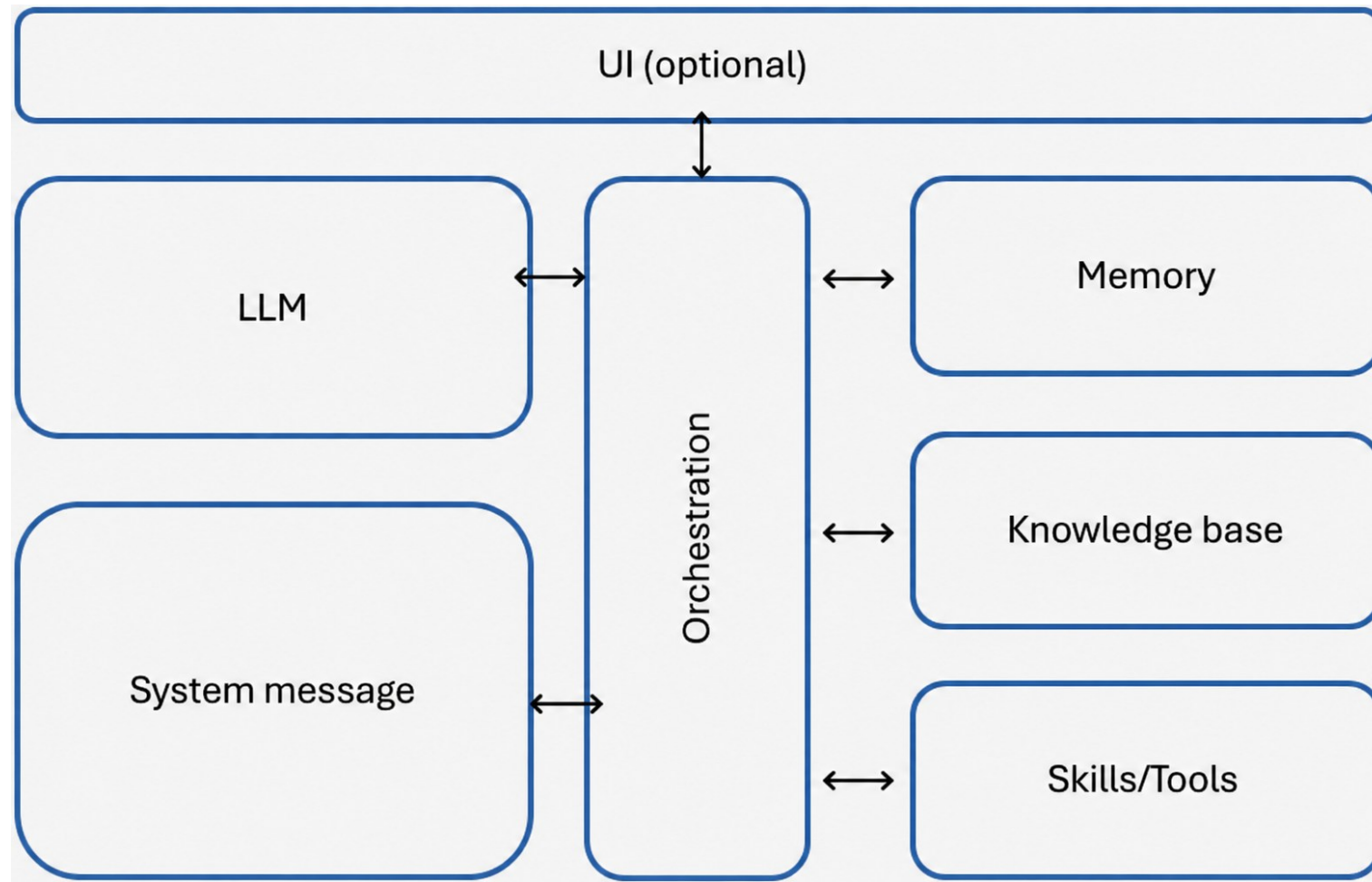
But .. What are AI Agents?

AI Agents combine two aspects:

1. Intelligent entity capable of active decision-making and continuous learning.
2. Automated application or software designed to perform tasks on behalf of a user.



Anatomy of AI Agents





Anatomy of AI Agents: LLMs & System Messages

- In AI agents, **LLMs** are the primary reasoning engine.
 - LLMs process natural language, understand context, generate plans, and decide which tools or knowledge to access based on the provided input.
- **System messages** provide the foundational instructions, persona, and constraints for the AI.
 - They defines "who" the agent is and the "rules of engagement" for how it should behave and solve problems.
 - Follow prompting formats in constructing system messages.



Anatomy of AI Agents: Memory & Knowledge Bases

- **Memory** allows the agent to store and retrieve information from past interactions. Memory could be:
 - Short-term memory includes context from the current conversation.
 - Long-term memory includes historical data and persists after execution.
- **Knowledge bases** provide a structured or unstructured repository of specific information (e.g., company documents or technical manuals).
 - Examples include Vector Stores, relational databases, Graph databases.

Anatomy of AI Agents: Orchestrators

- **Orchestrators** form the central "brain" of the system.
- Orchestrators manage the workflow, route information between the user, the LLM, and various support modules (memory, tools, etc.) to ensure the agent's goals are met in a logical sequence.
- Examples include LangGraph, CrewAI, Microsoft's AutoGen.





Anatomy of AI Agents: Tools

- **Tools** offer external functions or APIs that the agent can execute to perform actions in the physical or digital world.
- Tools can have different interfaces and agent can struggle to access Tools properly.
- The **Model Context Protocol (MCP)** was invented to standardize tool access and use among agents.
- Q. What are the tools you may use in an AI research agent?



Anatomy of AI Agents: Skills

- **Skills** are packaged expertise and instructions that teach an agent how to perform tasks consistently.
- Skills are typically embedded within the LLM prompt.
- What is risky about using skills?

Anatomy of AI Agents: Skills

What are the risks of using skills?

1. A **prompt** expresses an instruction.
 2. A **policy** defines a rule.
 3. A **control** enforces a boundary.
- These are not interchangeable.

Notice the example on the right:

“Do not approve transactions above this limit”
as a prompt instruction may guide the LLM but it does not create a dependable transaction control.

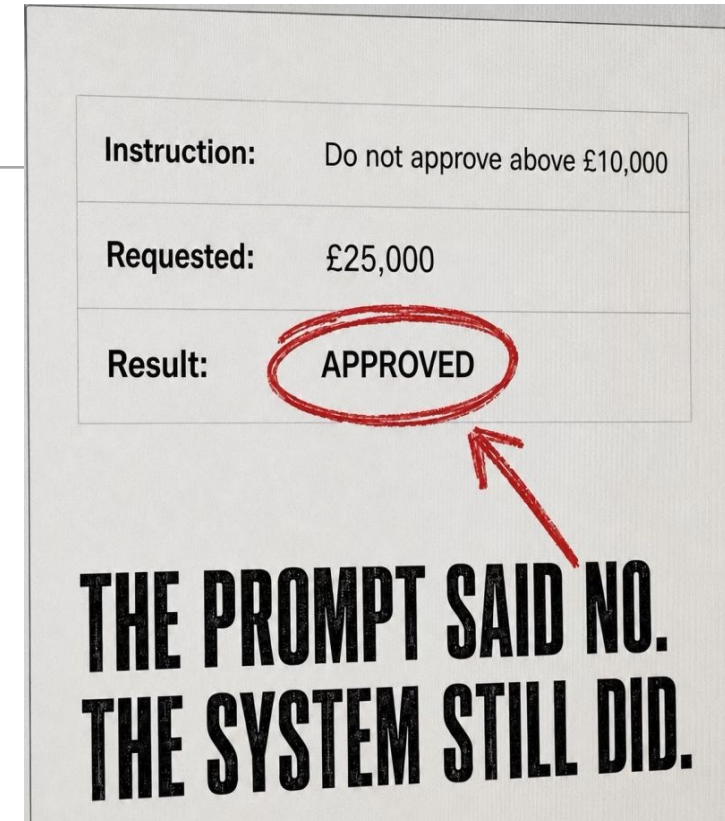


Image taken from Enterprise Intelligence Architecture by James Pearce

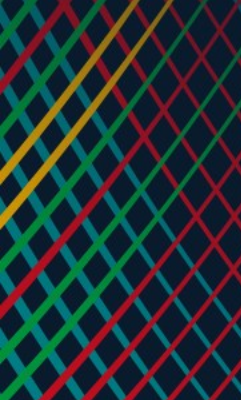


Anatomy of AI Agents: Skills

- Due to the unpredictable nature of LLMs, there have been several efforts to improve the context provided to LLMs to reduce the probability of hallucinations.
- Activities like building knowledge graphs of information (e.g., [Obsidian](#), [LLM Wiki](#), GraphRAG, etc.) and strengthening the AI agent harness have been popular in handling skill flaws.
- **However, these techniques don't solve the problem fundamentally.**
- **If you are looking for a 100% accurate output, you need a deterministic approach to evaluate the LLM outcome (or Human-In-The-Loop approach)**

Anatomy of AI Agents: Tools vs Skills

Feature	Tools	Skills
Setup Difficulty	Moderate (requires dev. knowledge)	Minimal (Markdown only)
Execution	Fixed-schema API calls	Natural-language instructions
Performance	Network latency	Local, no network overhead
Best for	Precise, predictable operations	Guidance & contextual adaptation
Maintenance	Centralized updates	Manual Markdown edits
Scalability	Tool discovery gets complex	Selection complexity grows
Potential Failures	Wrong tool selection	Misinterpretation + wrong selection



What are the Best Agentic Engineering IDEs?

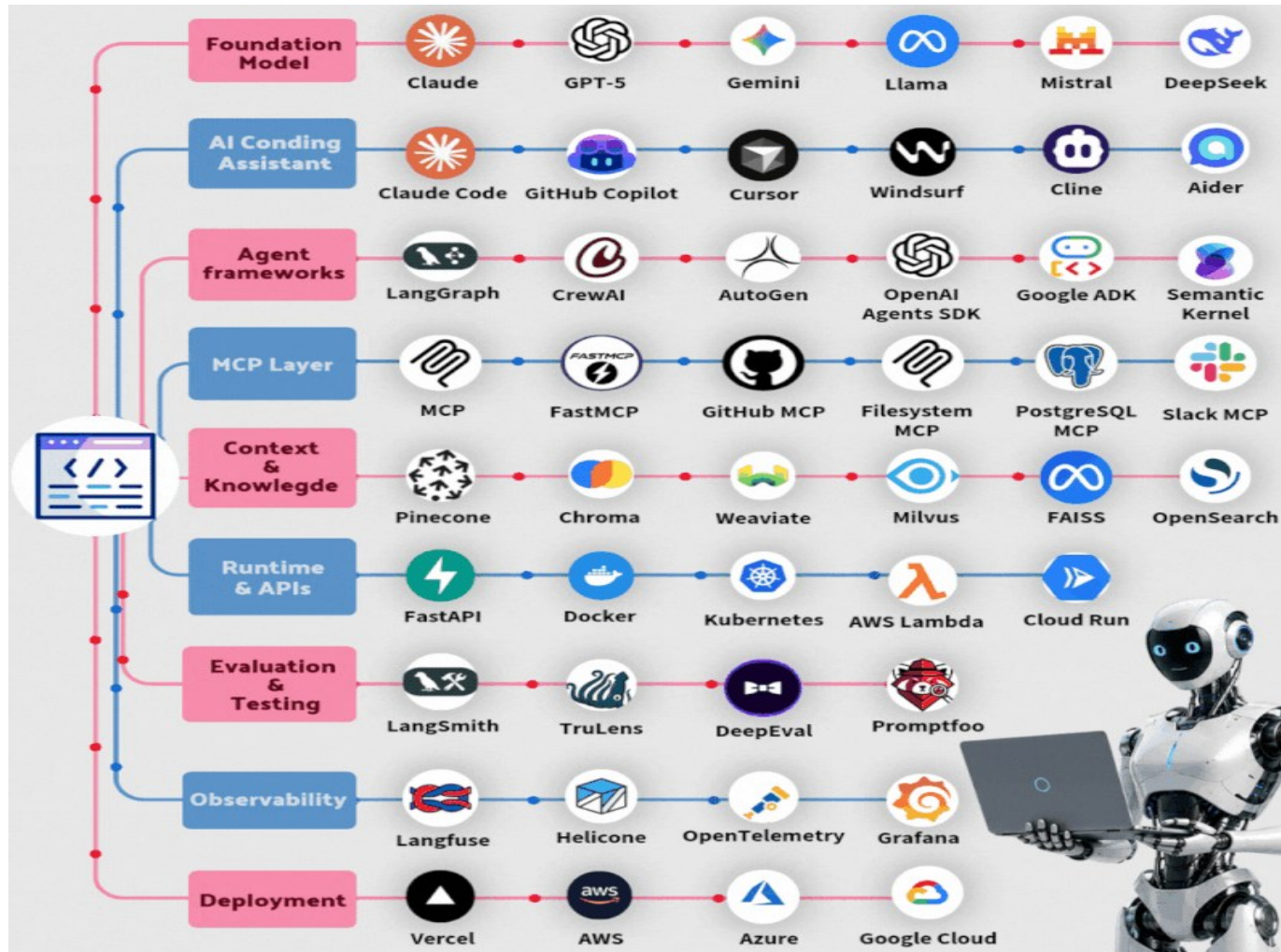
(As of January 2026)

Platform	Real-time completion	Multi-file context aware	NL to code generation	Explain docs/ comments	Refactoring suggestions	Test generation	Security vulnerability detection	Cloud option	On-prem/self-host option	BYOK models	Open-weight models	Proprietary models	Plugin/IDE integration	Standalone IDE/editor	Free tier	Offline capability	Broad language support	Low-latency completion
Google Antigravity	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Cursor	Yes	Yes	Yes	Yes	Yes	-	-	Yes	Yes	Yes	-	Yes	No	Yes	Yes	No	-	Yes
VS Code (with AI extensions)	Yes	Yes	Yes	Yes	Yes	Yes	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
GitHub Copilot	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Amazon Q Developer (Code Whisperer)	Yes	-	Yes	-	Yes	-	Yes	Yes	No	-	-	Yes	Yes	No	Yes	No	-	Yes
Tabnine	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	-	Yes	Yes	Yes
Replit Ghostwriter/Agent	Yes	Yes	Yes	Yes	Yes	-	Yes	Yes	No	Yes	-	Yes	No	Yes	Yes	No	-	Yes
JetBrains AI Assistant	Yes	Yes	Yes	Yes	Yes	Yes	-	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
Sourcegraph Cody	Yes	Yes	Yes	Yes	Yes	Yes	-	Yes	Yes	-	-	-	-	No	-	-	-	Yes
Continue	Yes	-	Yes	Yes	Yes	-	-	Yes	Yes	Yes	Yes	Yes	Yes	No	-	Yes	-	-

Additional Resources

- **Required Activities:** Watch the following videos and be ready to answer future questions related to them:
 - Coding agent harnesses
<https://www.youtube.com/watch?v=J07YhyzIGXw>
 - Personal AI Agents
<https://www.youtube.com/watch?v=gTsAGm0B0aw>
 - Skills vs Tools in AI Agents
<https://www.youtube.com/watch?v=xc16cRV3egg>
- **Optional Activity:** Do you have a GPU with 20GB VRAM and interested in building your personal AI agent?
 - Deploy your Personal AI Agent: <https://github.com/CMU-AGAI/democratized-personal-agent>

For Your Information



Waitlisted Students

